



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 1

- 1. Title:** Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.
- 3. Theory:** JavaScript is a scripting language used to make web pages interactive. It can be added in three ways:
 - Inline JavaScript: Written inside HTML elements.
 - Internal JavaScript: Written inside the `<script>` tag in the HTML file.
 - External JavaScript: Written in a separate `.js` file and linked to the HTML page.
 - Console methods like `console.log()`, `console.info()`, `console.warn()`, and `console.error()` are used for debugging and displaying messages.



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4. Output:

file:///C:/Users/HP/Desktop/sharvariw(1207)index.html

MySpace

Hello, I'm Sharvari

Welcome to my little corner of the internet. I love technology, creativity, and building amazing things. I am a passionate learner with a curiosity for technology, creativity, and innovation. I enjoy designing clean, modern websites, solving real-world problems through code, and continuously exploring new skills. This website gives you a glimpse into my journey, my interests, and the work I'm excited to create.

Know More

C:/Users/HP/Desktop/myportfolio/index.html#about

MySpace

Home About

My Skills

HTML

CSS

JavaScript

Python

Java

Problem Solving

Contact

Email : sharvari@gmail.com

LinkedIn : linkedin.com/in/sharvari

Change Footer Show Contact Show Date & Time

5. **Result/Conclusion:** The experiment was performed successfully. A webpage displaying a welcome message and user information was created using inline, internal, and external JavaScript, and console methods were demonstrated



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Case Study No.: 1

1. **Case Study Title:** Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</title>
  <style>
    body {
      font-family: 'Times New Roman', Times, serif;
      background-color: #FFB6C1;
      text-align: center;
      margin: 0;
      padding: 0;
    }
    h1 {
      color: brown;
    }
    .welcome-msg {
      font-size: 1.5rem;
      color: #333;
      margin: 20px 0;
      font-weight: bold;
    }
    .form-container {
      margin-top: 20px;
    }
    input {
      margin: 5px;
      padding: 8px;
      width: 250px;
    }
    button {
      padding: 8px 15px;
      background-color: brown;
      color: white;
      border: none;
      cursor: pointer;
      font-weight: bold;
    }
    button:hover {
      background-color: #6e1a1a;
    }
  </style>
</head>
<body>
  <h1>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</h1>
  <div class="welcome-msg">
    <p>WELCOME TO SIT NAGPUR</p>
  </div>
  <div class="form-container">
    <input type="text" value="Student Information Page" />
    <button>Link</button>
  </div>
</body>
</html>

/* --- CUSTOM POPUP MODAL STYLES --- */
```



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```
.modal-overlay {
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background-color: rgba(0, 0, 0, 0.5); /* Dims the background */
  display: flex;
  justify-content: center;
  align-items: center;
  z-index: 1000;
}

.modal-box {
  background-color: white;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  text-align: center;
  max-width: 400px;
  width: 90%;
}

.modal-box h4 {
  margin-top: 0;
  color: brown;
  font-size: 1.3rem;
}

.modal-buttons {
  margin-top: 20px;
  display: flex;
  justify-content: center;
  gap: 15px;
}

.btn-yes {
  background-color: #28a745;
}

.btn-no {
  background-color: #dc3545;
}
</style>
</head>
<body>

<!-- CUSTOM POPUP MODAL -->
<div id="customModal" class="modal-overlay">
  <div class="modal-box">
    <h4>Registration Check</h4>
    <p id="modalQuestion">Are you already registered with us?</p>
    <div class="modal-buttons" id="actionButtons">
      <!-- User clicks YES -->
      <button class="btn-yes" onclick="handleUserChoice(true)">Yes, I am</button>
      <!-- User clicks NO -->
      <button class="btn-no" onclick="handleUserChoice(false)">No, I'm new</button>
    </div>
  </div>
</div>
```



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<!-- MAIN BACKGROUND CONTENT -->

<header>

<h1>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</h1>

<p class="welcome-msg">Welcome to the Student Portal!</p>

<hr>

<h2>Department Of Computer Science and Engineering</h2>

<h3>Specialization: AI & ML / Data Science</h3>

</header>

<main class="form-container">

<h3>Student Information Form</h3>

<input type="text" id="nameInput" placeholder="Enter Full Name">

<input type="text" id="rollInput" placeholder="Enter Roll Number">

<input type="email" id="emailInput" placeholder="Enter Email Address">

<button onclick="collectData()">Submit Data</button>

</main>

<script>

// Handles what happens after clicking Yes or No

function handleUserChoice(isRegistered) {

const questionText = document.getElementById("modalQuestion");

const buttonsContainer = document.getElementById("actionButtons");

// 1. Change the text to display the "Welcome to SIT" message

if (isRegistered) {

questionText.innerHTML = "Welcome to SIT!
Please proceed to login.";

} else {

questionText.innerHTML = "Welcome to SIT!
Please create an account on our registration page.";

}

// 2. Replace the Yes/No buttons with a single "OK / Close" button to avoid browser blocks

buttonsContainer.innerHTML = '<button onclick="closeModal()">OK</button>';

}

// Closes the modal box completely to reveal the form

function closeModal() {

document.getElementById("customModal").style.display = "none";

}

// Logic for collecting form data

function collectData() {

const name = document.getElementById("nameInput").value;

const rollNo = document.getElementById("rollInput").value;

const email = document.getElementById("emailInput").value;

if (!name || !rollNo || !email) {

alert("Please fill out all fields.");

return;

}

localStorage.setItem("studentName", name);

localStorage.setItem("studentRoll", rollNo);

localStorage.setItem("studentEmail", email);

window.location.href = "student.html";

}



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```
</script>
</body>
</html>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Information - SIT</title>
  <style>
    body {
      font-family: 'Times New Roman', Times, serif;
      background-color: #FFB6C1;
      text-align: center;
      padding: 20px;
    }
    h1 {
      color: brown;
    }
    .info-card {
      background-color: white;
      max-width: 400px;
      margin: 30px auto;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0px 4px 8px rgba(0,0,0,0.1);
      text-align: left;
    }
    .info-item {
      margin-bottom: 15px;
      font-size: 1.1rem;
    }
    .label {
      font-weight: bold;
      color: brown;
    }
    a {
      display: inline-block;
      margin-top: 20px;
      color: brown;
      text-decoration: none;
      font-weight: bold;
    }
  </style>
</head>
<body>

  <h1>SIT Student Profile</h1>
  <h2>Submitted Information</h2>

  <div class="info-card">
    <div class="info-item"><span class="label">Name:</span> <span id="displayName"></span></div>
    <div class="info-item"><span class="label">Roll Number:</span> <span id="displayRoll"></span></div>
    <div class="info-item"><span class="label">Email:</span> <span id="displayEmail"></span></div>
  </div>
```



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[< Go Back to Form](index.html)

<script>

// Fetch the stored data on page load

window.onload = function() {

const name = localStorage.getItem("studentName");

const rollNo = localStorage.getItem("studentRoll");

const email = localStorage.getItem("studentEmail");

// Display data, or show a fallback message if empty

document.getElementById("displayName").innerText = name ? name : "No data found";

document.getElementById("displayRoll").innerText = rollNo ? rollNo : "No data found";

document.getElementById("displayEmail").innerText = email ? email : "No data found";

}

</script>

</body>



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Output:

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1index.html

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Welcome to the Student Portal!

Department Of Computer Science and Engineering

Specialization: AI & ML / Data Science

Student Information Form

Registration Check

Are you already registered with us?

Yes, I am

No, I'm new

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1index.html

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Welcome to the Student Portal!

Department Of Computer Science and Engineering

Specialization: AI & ML / Data Science

Student Information Form

Robert

1207

robertiscool@gmail.com

Submit Data



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1student.html

SIT Student Profile

Submitted Information

Name: robert

Roll Number: 1207

Email: robertiscool@gmail.com

[← Go Back to Form](#)

3. **Result/Conclusion:** The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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● Experiment No.: 2

1. Title: Use `var`, `let`, `const`, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

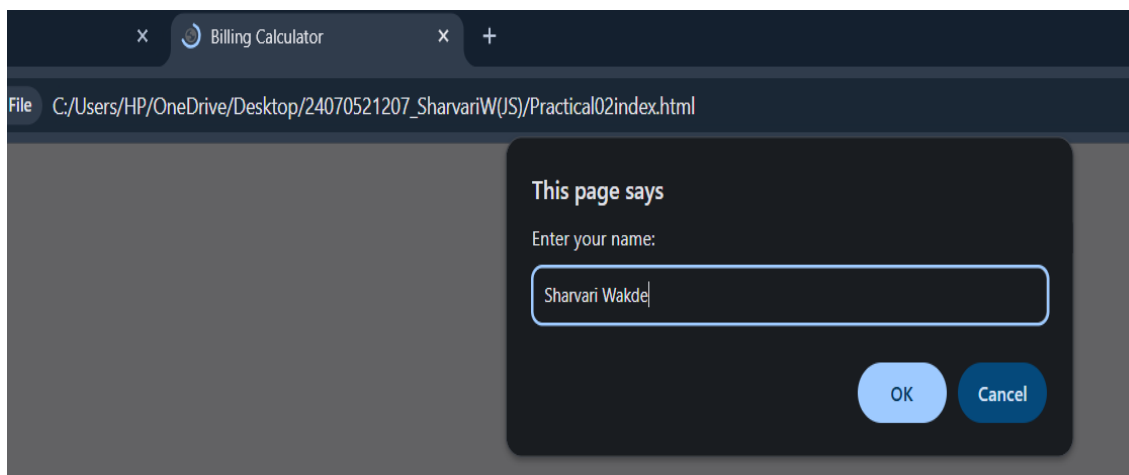
The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$ { }`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :





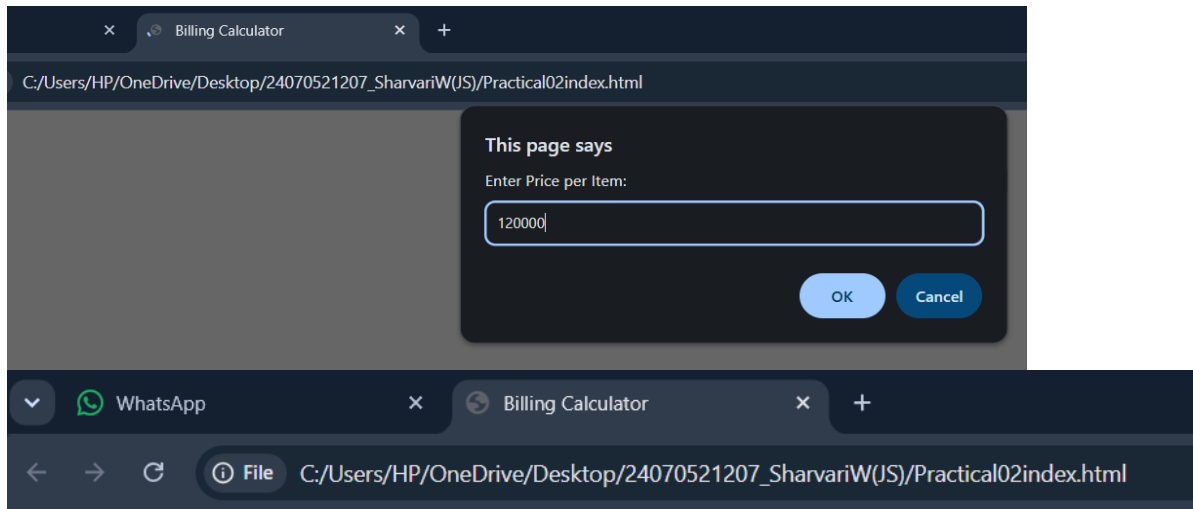
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Billing Calculator

Billing Receipt

Customer: Sharvari Wakde

Product: laptop

Quantity: 3

Price per Item: 120000

Total: 360000

GST (18%): 64800.00

Final Bill: 424800.00

5. **Result/Conclusion:** A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

- Case Study Title:** Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

2. Case Study Program Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Online Shopping Billing System</title>


<style>
  body{
    font-family: Arial, sans-serif;
    background:#f2f2f2;
  }

  .container{
    width:600px;
    margin:30px auto;
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0px 0px 10px gray;
  }

  h2{
    text-align:center;
    color:#0066cc;
  }

  table{
    width:100%;
  }

  td{
    padding:8px;
```



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}

```
input,select{
  width:100%;
  padding:8px;
}
```

```
button{
  width:100%;
  padding:10px;
  background:#0066cc;
  color:white;
  border:none;
  font-size:16px;
  cursor:pointer;
  border-radius:5px;
}
```

```
button:hover{
  background:#004c99;
}
```

```
#result{
  margin-top:20px;
  background:#e8f5e9;
  padding:15px;
  border-radius:5px;
  font-size:18px;
}
```

</style>

</head>

<body>

<div class="container">

<h2>ONLINE SHOPPING BILL</h2>

<table>

<tr>

<td>Customer Name</td>



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<td><input type="text" id="cname"></td>

</tr>

<tr>

<td>Mobile Number</td>

<td><input type="text" id="mobile"></td>

</tr>

<tr>

<td>Invoice Number</td>

<td><input type="text" id="invoice"></td>

</tr>

<tr>

<td>Product Name</td>

<td><input type="text" id="product"></td>

</tr>

<tr>

<td>Quantity (Kg)</td>

<td><input type="number" id="qty"></td>

</tr>

<tr>

<td>Rate per Kg</td>

<td><input type="number" id="rate"></td>

</tr>

<tr>

<td>Discount (%)</td>

<td><input type="number" id="discount"></td>

</tr>

<tr>

<td>GST (%)</td>

<td><input type="number" id="gst"></td>

</tr>

<tr>

<td>Packing Charges</td>

<td><input type="number" id="packing"></td>

</tr>



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```
<tr>
<td>Payment Mode</td>
<td>
<select id="payment">
<option>Cash</option>
<option>Card</option>
<option>UPI</option>
</select>
</td>
</tr>

<tr>
<td>Membership</td>
<td>
<select id="member">
<option>Yes</option>
<option>No</option>
</select>
</td>
</tr>

<tr>
<td colspan="2">
<button onclick="calculateBill()">Generate Bill</button>
</td>
</tr>

</table>

<div id="result"></div>

</div>

<script>

function calculateBill()
{

var cname=document.getElementById("cname").value;
var mobile=document.getElementById("mobile").value;
var invoice=document.getElementById("invoice").value;
var product=document.getElementById("product").value;
```



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```
var qty=parseFloat(document.getElementById("qty").value);
var rate=parseFloat(document.getElementById("rate").value);
var discount=parseFloat(document.getElementById("discount").value);
var gst=parseFloat(document.getElementById("gst").value);
var packing=parseFloat(document.getElementById("packing").value);
```

```
var payment=document.getElementById("payment").value;
var member=document.getElementById("member").value;
```

```
var amount=qty*rate;
```

```
var discountAmount=(amount*discount)/100;
```

```
var subtotal=amount-discountAmount;
```

```
var gstAmount=(subtotal*gst)/100;
```

```
var total=subtotal+gstAmount+packing;
```

```
if(member=="Yes")
{
    total=total-(total*5/100);
}
```

```
document.getElementById("result").innerHTML=
```

```
"<h3>Invoice</h3>"+
```

```
"<b>Customer Name :</b> "+cname+"<br><br>"+
```

```
"<b>Mobile No :</b> "+mobile+"<br><br>"+
```

```
"<b>Invoice No :</b> "+invoice+"<br><br>"+
```

```
"<b>Product :</b> "+product+"<br><br>"+
```

```
"<b>Quantity :</b> "+qty+" Kg<br><br>"+
```

```
"<b>Rate :</b> ₹"+rate+" /Kg<br><br>"+
```

```
"<b>Amount :</b> ₹"+amount.toFixed(2)+"<br><br>"+
```

```
"<b>Discount :</b> ₹"+discountAmount.toFixed(2)+"<br><br>"+
```



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```
"<b>GST :</b> ₹"+gstAmount.toFixed(2)+"<br><br>" +  
  
"<b>Packing Charges :</b> ₹"+packing.toFixed(2)+"<br><br>" +  
  
"<b>Payment Mode :</b> "+payment+"<br><br>" +  
  
"<b>Membership :</b> "+member+"<br><br>" +  
  
"<h2>Total Bill : ₹"+total.toFixed(2)+"</h2>";  
  
}  
  
</script>  
  
</body>  
</htm
```

Output:

file:///C:/Users/HP/Desktop/sharvariw(1207)/casestudy2.html

ONLINE SHOPPING BILL	
Customer Name	Sharvari
Mobile Number	456789765
Invoice Number	23000
Product Name	mobile
Quantity (Kg)	1
Rate per Kg	230988
Discount (%)	14
GST (%)	18
Packing Charges	200
Payment Mode	Cash
Membership	Yes
Generate Bill	



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file:///C:/Users/HP/Desktop/sharvariw(1207)/casestudy2.html

Invoice

Customer Name : Sharvari

Mobile No : 456789765

Invoice No : 23000

Product : mobile

Quantity : 1 Kg

Rate : ₹230988 /Kg

Amount : ₹230988.00

Discount : ₹32338.32

GST : ₹35756.94

Packing Charges : ₹200.00

Payment Mode : Cash

Membership : Yes

Total Bill : ₹222876.29

Result/Conclusion: The online shopping billing webpage was successfully developed using **HTML, CSS, and JavaScript**. The program accepted customer and product details, calculated the total bill including **discount, GST, and packing charges**, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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● Experiment No.: 3

1. **Title:** Implement control structures and form validation; create a grading system based on user-entered marks.

2. **Software/Tools Required:** VS Code, web browser, HTML, JavaScript.

3. **Theory: Control structures** (if, else if, and else) are used to make decisions based on the marks entered by the user.

Form validation checks whether the input is empty or outside the valid range (0–100) before processing.

The `getElementById()` method is used to access the value entered in the input box.

The `Number()` function converts the entered value from text to a numeric value. A **function** (`calculateGrade()`) is called when the **Submit** button is clicked to validate the input and calculate the grade.

The calculated grade is displayed on the webpage using the `innerHTML` property.

4. Output :

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

 **Sharvari's Grading System**

Enter Marks:

Submit

Marks: 95
Grade: A Grade



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says
Please enter marks!

OK

 **Sharvari's Grading System**

Enter Marks:

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says
Please enter valid marks between 0 and 100.

OK

 **Sharvari's Grading System**

Enter Marks:

5. Result/Conclusion:The grading system was successfully implemented using JavaScript. The program validated the user input, assigned grades using if-else control structures, and displayed the correct grade based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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Case Study No.: 3

- 1. Case Study Title:** Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Write a JavaScript program that accepts student marks and display the Scholarship category.

Case Study Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Scholarship Eligibility</title>
```

```
<style>
```

```
body {
```

```
margin: 0;
```

```
font-family: Arial, sans-serif;
```

```
background: linear-gradient(135deg, hsl(21, 75%, 69%), hsl(39, 52%, 79%));
```

```
display: flex;
```

```
justify-content: center;
```

```
align-items: center;
```

```
height: 100vh;
```



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}

.container {

background: white;

padding: 35px;

border-radius: 15px;

box-shadow: 0 8px 20px rgba(0,0,0,0.2);

text-align: center;

width: 350px;

}

h2 {

color: #333;

margin-bottom: 20px;

}

input {

width: 90%;

padding: 10px;

margin-top: 10px;

border: 2px solid #ccc;

border-radius: 8px;



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font-size: 16px;

}

button {

margin-top: 15px;

padding: 10px 25px;

background: hsl(32, 39%, 49%);

color: white;

border: none;

border-radius: 8px;

font-size: 16px;

cursor: pointer;

transition: 0.3s;

}

button:hover {

background: hsl(60, 100%, 41%);

}

#result {

margin-top: 20px;

font-size: 20px;



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font-weight: bold;

color: #333;

}

</style>

</head>

<body>

<div class="container">

<h2>🎓 Sharvari's Scholarship Eligibility System [24070521207] </h2>

<label>Enter Marks:</label>

<input type="number" id="marks" placeholder="Enter marks (0-100)">

<button onclick="checkGrade()">Submit</button>

<p id="result"></p>

</div>

<script>

function checkGrade() {

var marks = Number(document.getElementById("marks").value);



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```
var input = document.getElementById("marks").value;
```

```
if (input === "") {
```

```
    alert("Please enter marks!");
```

```
    return;
```

```
}
```

```
if (marks < 0 || marks > 100) {
```

```
    alert("Please enter valid marks between 0 and 100.");
```

```
    return;
```

```
}
```

```
// Declared the scholarship variable properly
```

```
var scholarship = "";
```

```
if (marks >= 95) {
```

```
    scholarship = "Platinum Scholarship";
```

```
} else if (marks >= 85) {
```

```
    scholarship = "Gold Scholarship";
```



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```
} else if (marks >= 70) {  
  
    scholarship = "Silver Scholarship";  
  
} else if (marks >= 50) {  
  
    scholarship = "Bronze Scholarship";  
  
} else {  
  
    // Simplified the final fallback case  
  
    scholarship = "Not Eligible for Scholarship";  
  
}  
  
// Now variables match perfectly  
  
document.getElementById("result").innerHTML = "Marks: " + marks +  
"<br>Scholarship: " + scholarship;  
  
}  
  
</script>  
  
</body>  
  
</html>
```

```
marks = Number(marks);
if(marks < 0 || marks > 100)
{
    alert("Enter Valid Marks for Eligibility");
    return;
}
let grade;
if(marks >= 95)
    grade = "Platinum Scholarship";
else if(marks >= 85)
    grade = "Gold Scholarship";
else if(marks >= 70)
    grade = "Silver Scholarship";
else if(marks >=50)
    grade = "Bronze Scholarship";
else if(marks <= 50)
    grade = "Not Eligible for Scholarship";
document.getElementById("result").innerHTML =
    "Marks: " + marks + "<br>Grade: " + grade;
}
</script>
</body>
</html>
```



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3. Output:

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

Sharvari's Scholarship Eligibility System [24070521207]

Enter Marks:

Submit

Marks: 90
Scholarship: Gold Scholarship

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter valid marks between 0 and 100.

OK

Sharvari's Scholarship Eligibility System [24070521207]



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter marks!

OK



**Sharvari's Scholarship
Eligibility System
[24070521207]**

Enter Marks:

Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web developmen



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● Experiment No.: 4

1. Title: Use functions types, scope, and closures; apply try-catch; build palindrome checker.

2. Software/Tools Required: VS Code, web browser, HTML, JavaScript.

3. Theory:

1. Regular Function

A regular function is a reusable block of code that performs a specific task. It can accept parameters, return values, and is invoked whenever needed. Functions improve code reusability and organization.

2. Function Declaration

A function declaration defines a function using the function keyword and a function name. It is hoisted, meaning it can be called before its declaration in the code.

3. Function Expression

A function expression stores a function inside a variable. Unlike function declarations, it is not fully hoisted and can only be called after it is defined.

4. Anonymous Function

An anonymous function is a function without a name. It is commonly used as a callback function or assigned to a variable for one-time or temporary operations.

5. Arrow Function

An arrow function is a shorter syntax for writing functions using the => operator. It is concise, easier to read, and automatically inherits the this value from its surrounding scope.

6. Variable Scope

Variable scope determines where a variable can be accessed in a program. JavaScript provides global scope, function scope, and block scope (using let and const).



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4. Output :

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✓ **"2442" is a Palindrome**

 Checks Performed : 1

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✗ **"2449" is NOT a Palindrome**

 Checks Performed : 2

5. Result/Conclusion: The JavaScript concepts such as functions, scope, closures, arrow functions, and exception handling were successfully studied and implemented. Their behaviour and usage were verified through simple programs. JavaScript functions and scope mechanisms help in writing clean, reusable, and efficient code. Understanding these concepts improves programming skills and enables the development of reliable and maintainable web app



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Case Study No.: 4

1. **Case Study Title: ATM system verifies if pin is palindrome. if palindrome,mesplay security message . to implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaraction,function expression ,arrow function and demonstrate scope and closure concepts.**

Case Study Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>ATM PIN Verification</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
background-color: #f5f0eb; /* Soft beige background */
```

```
color: #4a3c31; /* Deep dark brown text */
```

```
display: flex;
```

```
flex-direction: column;
```

```
align-items: center;
```

```
justify-content: center;
```



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height: 100vh;

margin: 0

}

.card {

background-color: #ffffff;

padding: 30px;

border-radius: 10px;

box-shadow: 0 4px 10px rgba(74, 60, 49, 0.15);

text-align: center;

border-top: 5px solid #6b4c3b; /* Medium warm brown accent */

}

h2 {

color: #5c3a21; /* Rich chocolate brown */

}

label {

display: block;

margin-bottom: 10px;

font-weight: bold;

}



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/* Changed selector from input[type="password"] to input[type="text"] */

input[type="text"] {

padding: 8px;

width: 200px;

border: 1px solid #b8a398;

border-radius: 4px;

margin-bottom: 15px;

}

button {

background-color: #6b4c3b; /* Warm brown button */

color: white;

padding: 9px 18px;

border: none;

border-radius: 4px;

cursor: pointer;

font-weight: bold;

}

button:hover {

background-color: #5c3a21;



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```
}  
  
#result {  
  
    margin-top: 20px;  
  
    font-size: 14px;  
  
}  
  
</style>  
  
</head>  
  
<body>  
  
<div class="card">  
  
    <h2>Sharvari's (1207) PIN Verification</h2>  
  
    <label for="pin">Enter ATM PIN:</label>  
  
    <input type="text" id="pin">  
  
    <button onclick="verifyPIN()">Verify</button>  
  
    <h3 id="result"></h3>  
  
</div>  
  
<script>
```



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// Function Declaration - defines a function with the function keyword and a function name (use when main part of program and you want it available thruout the scope)

```
function reversePIN(pin) {  
  
    return pin.split("").reverse().join("");  
  
}
```

// Function Expression - stores a function inside the variable (use when passing functions as arguments, assigning functions to variables , creating callbacks)

```
const isPalindrome = function(pin) {  
  
    return pin === reversePIN(pin);  
  
};
```

// Arrow Function - shorter way to write a function in js. it was introduced in ES6 and uses arrow syntax

```
const displayMessage = (msg) => {  
  
    document.getElementById("result").innerHTML = msg;  
  
};
```

// Scope and Closure

```
function verifyPIN() {
```



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```
let pin = document.getElementById("pin").value;

function securityMessage() {

    if (isPalindrome(pin)) {

        return " PIN is a Palindrome. Hence, Its a weak password. Please Change it for
max security";

    } else {

        return " PIN is Not a Palindrome. So, security verified!";

    }

}

displayMessage(securityMessage());

}

</script>

</body>

</html>
```



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Output:



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is a Palindrome. Hence, Its a weak password. Please Change it for max security



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is Not a Palindrome. So, security verified!



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3. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.